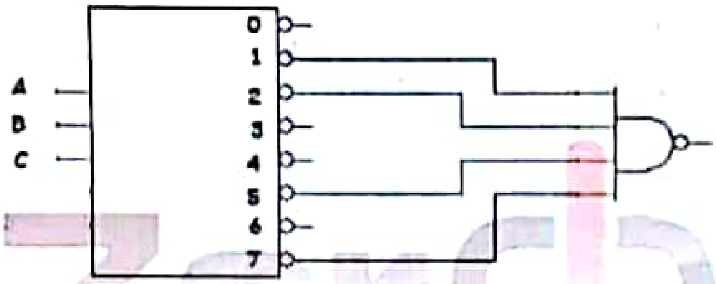
Al-Balqa Applied University Faculty of Engineering Technology Dept. of Computer Engineering **Second Exam** *Date:* 9/4/2016	2015-2016	جامعة البلقاء التطبيقية كلية الهندسة التكنولوجية قسم هندسة الحاسبات اسم الطالب: الرقم الجامعي: الوقت:

Select the correct answer, from options (a,b,c,d) to the following questions. (5marks)

	1	2	3	4	5
Ans.	a	c	d	a	d

A D A D

1. The function that can be implemented from the following decoder is:
F(A,B,C) = ...



AND → π

a- ∑(1,2,5,7). b- π(1,2,5,7). c- ∑(0,3,4,6). d- a and b

BCD 2. How many 4-bit parallel adders would be required to add two binary numbers each representing decimal numbers up to three hexadecimal digits?
a- 1 b- 2 c- 3 d- 4

3. You can implement the overflow bit using
a- AND gate b- OR gate c- XOR gate d- XNOR gate

4. To multiply two binary numbers, the multiplier = 4 by the multiplicand = 5 , You will need:
a- 20 AND gates b- 9 AND gates c- 15 AND gates d- 16 AND gates

5. To multiply two binary numbers, the multiplier = 4 by the multiplicand = 5 , You will need:
a- Two 4-bit adders
b- Two 5-bit adders
c- ~~Three 4-bit adders~~
d- Three 5-bit adders

A: $A_3\ A_2\ A_1\ A_0$
B: $B_4\ B_3\ B_2\ B_1\ B_0$

$A_0B_4\ A_0B_3\ A_0B_2\ A_0B_1\ A_0B_0$
$A_1B_4\ A_1B_3\ A_1B_2\ A_1B_1\ A_1B_0$
$A_2B_4\ A_2B_3\ A_2B_2\ A_2B_1\ A_2B_0$
$A_3B_4\ A_3B_3\ A_3B_2\ A_3B_1\ A_3B_0$

BCD

Q2) Design a 4-bit adder which implement the following conditions.

A- If the addition result is >= 12 or a carry out then number 3 must be added to the result, Else add zero.

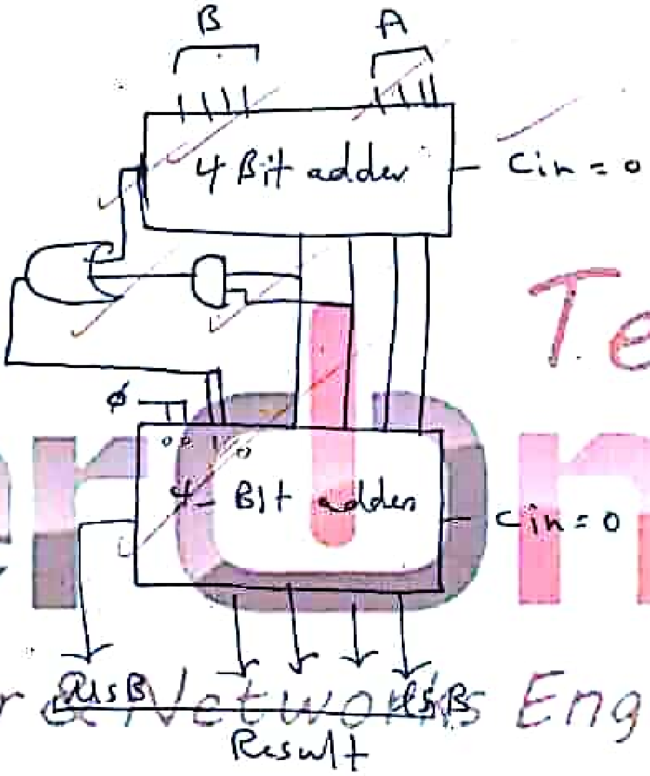
Q3) Implement the function FABCD=∑(0,3,6,7,8,11,14,15) using Multiplexer, Use inputs A,C,D as select lines.

Q4) Design the following function: $F(A,B,C,D) = (\bar{A}B + AC)\bar{D}$ using:

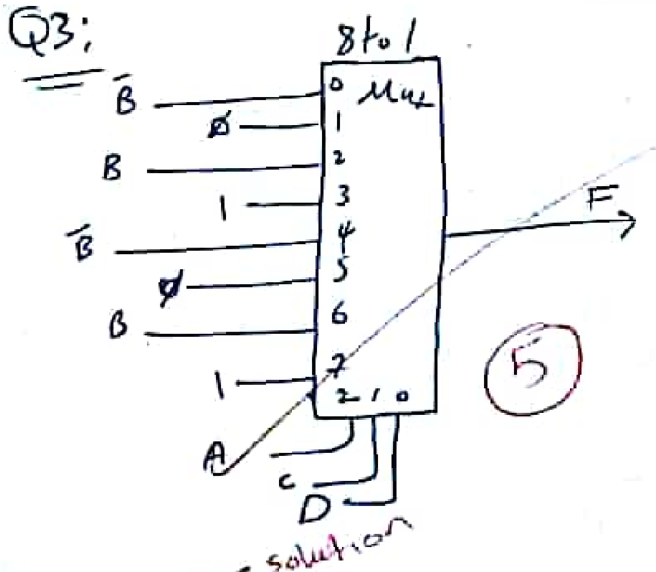
a) NAND gates only.

b) NOR gates only.

Q2:



Q3:



000 → B̄ → 1

0011 → ∅ → 1

Q4: F = Σ(4, 6, 10, 14)

a) Nand → Sop

F

← solution

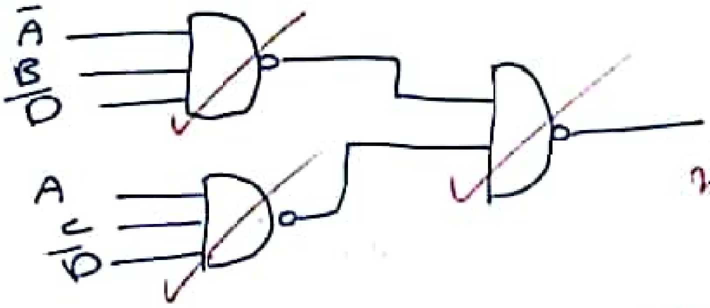
Nand gates → SOP

$F = \Sigma(4, 6, 10, 14)$

$F = \bar{A} B \bar{D} + A C \bar{D}$

$F = \overline{\overline{\bar{A} B \bar{D}} \cdot \overline{A C \bar{D}}}$

2.5

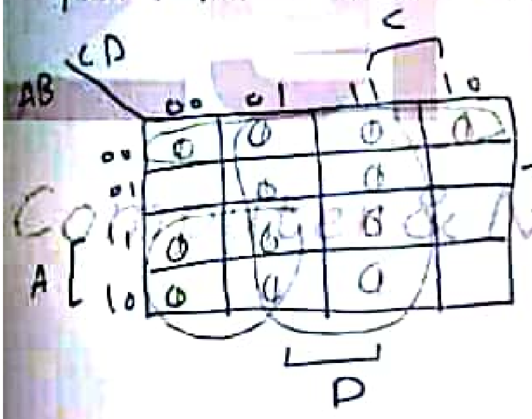


2.5

b) NOR Gates only :-

minterm
Maxterm : 0 : A
1 : $\bar{A}$

$F = \Pi(0, 1, 2, 3, 5, 7, 8, 9, 11, 12, 13, 15)$



~~$F = \bar{A}\bar{B} +$~~

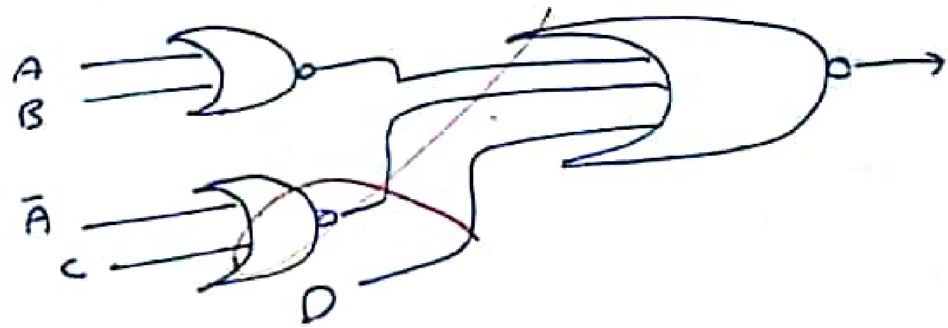
$F = (\bar{A} + \bar{B}) \cdot (D) \cdot (A + \bar{C})$

$F = (A + B) \cdot (\bar{D}) \cdot (\bar{A} + C)$

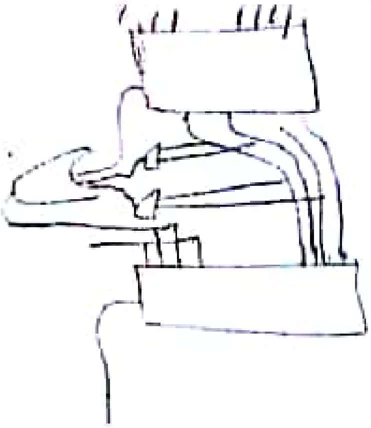
~~$F(0000) = 0 = (0+0) \cdot (0) \cdot (1+0) = 0$~~

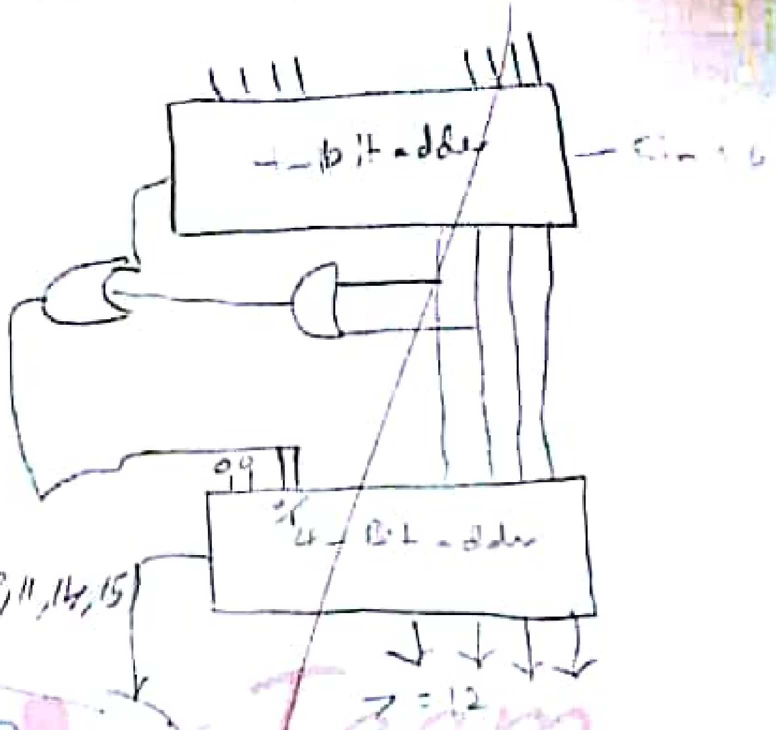
$F = \overline{\overline{(A+B) \cdot (\bar{D}) \cdot (\bar{A}+C)}}$

$F = \overline{\overline{(A+B)} + (D) + \overline{(\bar{A}+C)}}$



Addition (>=12) or a carry out then add the number?
else add zero





Q3: $F(A,B,C,D) = \Sigma(0,3,6,7,8,11,14,15)$

Q5) A C D → C.S

$\bar{B}$: 0, 1, 2, 3, 8, 9, 10, 11

B: 4, 5, 6, 7, 12, 13, 14, 15

$\bar{B}$ 0 B 1 $\bar{B}$ 0 B 1

	B	F
0	0	1
1	0	0
2	0	0
3	0	1
4	1	0
5	1	0
6	1	1
7	1	1
8	0	1
9	0	0
10	0	0
11	0	1
12	1	0
13	1	0
14	1	1
15	1	1

Q4: $F(A,B,C,D) = (\bar{A}B + AC)\bar{D}$

a) Nand gates.
b) Nor gates

$F = \bar{A}B\bar{D} + AC\bar{D}$

0100 1010

0110 1110

$F = \pi(0,1,2,3,5,$

$F = \Sigma(4,6,10,14)$

Nand → sop

$F = \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C\bar{D} + ABC\bar{D}$